1	Egg Storage Container	PP-C HI / Polypropylene Copolymer		1	Yield: ~22 MPa | Density: 0.90 g/cm³ | Temp: -20 to 100°C	1
2	Egg Tunnel	PP-C HI / Polypropylene Copolymer		1	Yield: ~22 MPa | Density: 0.90 g/cm³ | Temp: -20 to 100°C	2
3	Gates for Transportation Tunnel	Acetal Copolymer (POM-C)		2	Yield: ~62 MPa | Density: 1.41 g/cm³ | Temp: -40 to 100°C	4
4	Gate Shaft Holder for Tunnel	POM-C / Acetal Copolymer		4	Yield: ~62 MPa | Density: 1.41 g/cm³ | Temp: -40 to 100°C	5
5	Egg Tunnel Shaft	EN 1.4301 / AISI 304		4	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	6
6	Egg Holding Shaft	EN AW-6082 T6		1	Yield: ~260 MPa | Density: 2.70 g/cm³ | Temp: -40 to 150°C	7
7	Egg Holding Arm	EN AW-6082 T6		1	Yield: ~260 MPa | Density: 2.70 g/cm³ | Temp: -40 to 150°C	8
8	Base of Spear	EN AW-6082 T6 (clear anodised)		1	Yield: ~260 MPa | Density: 2.70 g/cm³ | Temp: -40 to 150°C	9
9	Spears (Left and Right)	EN 1.4116 / X50CrMoV15 (hardened)		2	Yield: ~800–1000 MPa | Density: 7.70 g/cm³ | Temp: -20 to 200°C	10
10	Solenoid Motor Mounting Bracket	EN 1.4301 / AISI 304		2	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	11
11	Solenoid Motor Casing	EN AW-6082 T6 (clear anodised)		3	Yield: ~260 MPa | Density: 2.70 g/cm³ | Temp: -40 to 150°C	12
12	Main Drawer Box	EN 1.4301 / X5CrNi18-10		1	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	13
13	Drawer Front Cover	ABS-HI plastic		1	Yield: ~40 MPa | Density: 1.05 g/cm³ | Temp: -20 to 80°C	14
14	Drawer Handle	ABS-HI plastic w/ brass inserts		1	Yield: ~40 MPa | Density: 1.05 g/cm³ | Temp: -20 to 80°C	15
15	Inner Shaft (Drawer)	EN 1.4301 / X5CrNi18-10		2	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	16
16	Manual Compression Plate	EN 1.4301 / X5CrNi18-10		1	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	17
17	Oil Storage Mount	EN AW-6082 T6		2	Yield: ~260 MPa | Density: 2.70 g/cm³ | Temp: -40 to 150°C	18
18	Pocket Plate	EN 1.4409 / GX2CrNiMo19-11-2 (cast)		4	Yield: ~185–205 MPa | Density: 7.90 g/cm³ | Temp: -196 to 400°C	19
19	Sprocket for Conveyor Belt	EN 1.4404 / X2CrNiMo17-12-2		4	Yield: ~205 MPa | Density: 8.00 g/cm³ | Temp: -196 to 400°C	20
20	Driving Shaft / Idler Shaft	EN 1.4404 / X2CrNiMo17-12-2		2	Yield: ~205 MPa | Density: 8.00 g/cm³ | Temp: -196 to 400°C	21
21	Conveyor Side Frame	EN 1.4301 / X5CrNi18-10		2	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	22
22	Conveyor Back Frame	EN 1.4301 / X5CrNi18-10		1	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	23
23	Conveyor Belt Hollow Support Frame	EN AW-6082 T6		1	Yield: ~260 MPa | Density: 2.70 g/cm³ | Temp: -40 to 150°C	24
24	Nema 17/Gearbox/Encoder Motor Housing	EN AC-43500 / AlSi10MnMg (cast)		1	Yield: ~140–180 MPa | Density: 2.68 g/cm³ | Temp: -40 to 150°C	25
25	Nema Motor Mount	EN 1.4301 / AISI 304		1	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	26
26	TOP IR Shielding	EN 1.4301 / X5CrNi18-10		1	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C (shield, near IR zone)	27
27	Ceramic Block for Holding Quartz IR			1	Flexural: ~300 MPa | Density: 3.70 g/cm³ | Temp: up to 1600°C	28
28	Side Lining for TOP IR	EN 1.4301 / X5CrNi18-10		2	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	29
29	Perforated Sheet	EN 1.4301 / X5CrNi18-10		1	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	30
30	TOP IR Back Frame	EN 1.4301 / X5CrNi18-10		1	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	31
31	Heating Block	EN AW-6082 T651		1	Yield: ~250 MPa | Density: 2.70 g/cm³ | Temp: -40 to 200°C	32
32	Side Frame (Bottom Heating)	EN 1.4301 / X5CrNi18-10		2	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	33
33	Front Frame (Bottom Heating)	EN 1.4301 / X5CrNi18-10		2	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	34
34	Bottom Frame (Bottom Heating)	EN 1.4301 / X5CrNi18-10		1	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	35
35	Mounting Brackets for Bottom Heating Block	EN 1.4301 / AISI 304		4	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	36
36	Aluminum Plate (Rotating Plate)	EN AW-6061 T6 (clear anodised)		1	Yield: ~275 MPa | Density: 2.70 g/cm³ | Temp: -40 to 150°C	37
37	Top Load Cell Plate	EN AW-6061 T6 (clear anodised)		1	Yield: ~275 MPa | Density: 2.70 g/cm³ | Temp: -40 to 150°C	38
38	Bottom Load Cell Plate	EN AW-6082 T6 (clear anodised)		1	Yield: ~260 MPa | Density: 2.70 g/cm³ | Temp: -40 to 150°C	39
39	Base Plate (Rotating Plate)	EN AW-6082 T6 (clear anodised)		1	Yield: ~260 MPa | Density: 2.70 g/cm³ | Temp: -40 to 150°C	40
40	Stepper Motor Housing	EN AC-43500 / AlSi10MnMg (cast, clear anodised)		1	Yield: ~140–180 MPa | Density: 2.68 g/cm³ | Temp: -40 to 150°C	41

Item no.	Part Name	Material specification	Link/Brand	Qty.	Technical Specification	Position in Technical assembly

faculty or department	technical reference	drawn by (last saved by)	drawing date	released by
Technology & Bionics		ahmad	09/12/2020	

HOCHSCHULE RHEIN-WAAL
Rhine Waal University of Applied Sciences

title, additional title	projection method	scale
SunnyPro Group3		

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drawing no. prefix	type of document
-ME.6.2701.03	bill of material

drawing no.	rev.	date of issue	sheet
260707-01-001	00	07/07/2026	1 /1

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41	Bottom Base for Rotating Plate	EN 1.4301 / AISI 304		1	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	42
42	Electrical Box Housing	EN 1.4301 / AISI 304		1	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	43
43	Main Frame Housing	EN 1.4301 / AISI 304		1	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	44
44	L Bracket for Servo Motor	EN 1.4301 / AISI 304		4	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	45
45	Plastic Back Plate (Display)	ABS-HI plastic w/ brass inserts		1	Yield: ~40 MPa | Density: 1.05 g/cm³ | Temp: -20 to 80°C	46
46	Front Frame for Casing (Display)	ABS-HI plastic		1	Yield: ~40 MPa | Density: 1.05 g/cm³ | Temp: -20 to 80°C	47
47	Main Case/Chassy Panels	ABS-HI plastic + purchased glass		1	Yield: ~40 MPa | Density: 1.05 g/cm³ | Temp: -20 to 80°C	48
48	DIN Rail	zinc-plated steel	Hutschiene TS 35 x 7,5mm gelocht - L=326mm	2	35 x 7,5 x 326 mm	49
49	Axle Shaft	Aluminum	Aluminum shaft	1	Ø 10 mm x100 mm	50
50	2-Bolt Flange Bearing Block	Aluminum	Stehlagereinheiten / Flanschlagereinheiten 2- Loch KFL000	4	Ø 10mm length 66mm total diameter 40 mm	51
51	Planetary Gearbox for Motor	Stainless Steel	Planetengetriebe Schrittmotor 27:1	1	Planetary gear 27:1	52
52	Countersunk Raised Head Cross Recess Screw	A2 Stainless Steel	Senkkopfschrauben M10x25	38	M10x25 / M10x60	53
53	Pan Head Cross Recess Screw	A2 Stainless Steel	Senkkopfschrauben M10x60	40	DIN 7985 (ISO 7045) / M8x16 / M3x12	54
54	Socket Head Cap Screw	A2 Stainless Steel	Linsenschrauben M8x16	12	M3x25 / M3x10mm	55
55	Pan Head Cross Recess Screw	A2 Stainless Steel	DIN 912/ISO 4762 A2 – M3 x 25 mm 100 Stück	26	ISO 7045 / M4x8	56
56	Prevailing Torque Hex Nut	Stainless Steel	DIN 912/ISO 4762 A2 – M3 x 10 mm 100 Stück	18	M6	57
57	Plain Washer	Polyoxymethylene	Flachkopfschraube ISO 7045 PZ M4x8-Z 4.8 galv.verz. 2000St.	10	M6	58
58	Socket Set Screw, Cup Point		Sechskantmuttern M6 DIN 934 Edelstahl A2	12	M5x20 / M5x30	59
59	Hex Thin Nut		DIN 439/ISO 4035 Sechskantmutter Edelstahl A2, M4 - 100 Stück	52	M3 / M4	60
60	Lazy Susan Bearing		Blickle 942239 DK-L-3 Stahlblech-Drehkranz 1 St.	2	140 x 110 mm	61
61	Rubber Drive Wheel		Freies Verschiffen 10PCS 8*32*16 flache typ PA nylon	3	8x32x16mm	62
62	IEC C20 Power Inlet		IEC 20 Power Inlet	1	16A, 250VAC	63
63	PCB Screw Terminal Connectors		20Paar 5.08mm Pitch Male Buchse PCB Screw Terminal Block | eBay.de	1	25x10x15mm, AC 300V, 15A	64
64	Axial Fan 8414NGML		EBM Papst 8414NGML Axiallüfter 24 V/DC 42 m³/h (L x B x H) 80 x 80 x 25mm | » Freue dich auf voelkner	2	80x80x25mm, 42m³/h, 24V DC	65
65	Gate Actuators JF-0530B		JF-0530B DC12V 300mA 5N/10mm Pull-Push-Type Solenoid Electromagnet	2	DC12V, 300mA, 5N/10mm, 30x16x15mm	66
66	Flip Actuator MG996R		MG996R Digital Servo Motor mit Metall Getriebe	1	0.17 sec/60° @ 4.8V, 40x19x43mm, 4.8-7.2V	67
67	Platform Solenoid JF-0837B		JF-0837B DC 24V 1A 15N 10mm Push Pull Type Open Frame Linear Motion Solenoid Electromagnet | Harfington	1	DC24V, 1A, 15N, 37x20x26mm	68
68	Peristaltic Pump		DC 12V Dosing Pump Peristaltic Pump	1	DC12V, Three roller, 11ml/min	69
69	Touchscreen Display Module - Nextion NX8048P070_011R		NEXTION 4.3/5.0/7.0" Intelligent LCD Touch Display	2	108x181x9.3mm	70
70	Stepper Motor		NEMA 17	1	2.8V, 5mm shaft, 2 phase	71
71	Incremental Rotary Encoder		AMT102-V	1	3.6V to 5.5V	72
72	Cartridge Heaters		Cartridge Heaters	4	250W, 230V AC	73
73	Temperature Sensor		PT100 temp. Sensor	1	-	74
74	Solid-State Relay		solid state relay 25A SSR-25AA , ZG3NC-325A Single phase	1	25A, 90-480VAC	75
75	Stepper Motor		FREI Standard-Schrittmotor, 5V	1	5V DC, 4 phase, 34.3 mN·m	76
76	Microswitch		5Pcs Mini Micro Limit Switch NO NC 3 Pins PCB Terminals SPDT 5A 125V 250V 29mm Roller Arm lever Snap Action Push Microswitches		5A, 125V/250V	77
77	Load Cell and HX711 Amplifier Module		Stemedu 5 kg Load Cell Digital Weight Sensor for Portable Electronic with HX711 Ad Amplifier Module, 11 lbs Weighing Sensor Arduino Raspberry Pi	1	4-wire, 5-10V, 12.7x12.7x80mm, voltage range 2.6-5.5V	78
Item no.	Part Name	Material specification	Link/Brand	Qty.	Technical Specification	Position in Technical assembly

faculty or department: Technology & Bionics	technical reference:	drawn by (last saved by): ahmad	drawing date: 09/12/2020	released by:	
HOCHSCHULE RHEIN-WAAL Rhine Waal University of Applied Sciences		title, additional title: SunnyPro Group 3		projection method	scale
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